The Theory of Actual Diplomatic Power

A Parabolic Model of International Diplomatic Representation

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Abstract

How much power does a diplomatic network *actually* generate? Foreign ministries are conventionally ranked by raw post counts—an input measure that ignores how network completeness converts presence into influence. This paper develops the *theory of actual diplomatic power*: international diplomatic representation (IDR), a state's posts normalized against the global leader, is mapped to actual diplomatic power (ADP) by a parabola, ADP = $\text{IDR}^2/100$. The quadratic is not arbitrary; we show it is the unique transform consistent with three weak axioms—scale-free normalization, homogeneity of degree two, and monotone continuity—and we defend convexity through the economics of networks: interaction channels compound pairwise, agenda-setting power is convex in coverage, and fixed costs of representation create scale economies. Applying the theory to all 66 entities of the Lowy Institute Global Diplomacy Index 2024 (7,745 posts), we find that the square law systematically amplifies concentration: the Gini coefficient of representation rises from 0.325 to 0.543, and a twelve-member "half-power club" commands 55.8% of global diplomatic power with only 36.9% of posts, while the median state, 2.56 times behind the leader in missions, falls 6.56 times behind in power. Rank–size regression recovers a power-law hierarchy (exponent −0.635), and mission counts scale with GDP (elasticity 0.32, $t = 11.1$) but not population, implying an ADP–GDP elasticity near 0.64. The parabolic lens turns headcounts into a power metric with sharp implications for middle powers, small states, and the European Union's twenty-two-fold coordination paradox.

The paper ends with "The End"

Keywords: diplomacy, diplomatic missions, power indices, concentration, power laws, international relations

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1 Introduction

How much power does a state's diplomatic network actually generate? Foreign ministries are customarily compared by the raw number of embassies and consulates they maintain: China leads with 274 posts, the United States follows with 271, and a long tail of small states trails far behind [7, 11]. Yet raw headcounts are an input measure, not a power measure. Intuitively, the two hundred and seventy-fourth post that completes a global network is worth more—not merely one unit more—than the twentieth post of a niche actor, because a complete network generates interactions, agenda-setting capacity, and crisis reach that a partial network cannot. Power, in other words, should be a *convex* function of representation.

This paper formalizes that intuition as *the theory of actual diplomatic power*. We define *International Diplomatic Representation* (IDR) as a state's stock of diplomatic posts normalized against the global leader, and we define *Actual Diplomatic Power* (ADP) as a parabolic function

of IDR: ADP $= (\text{IDR}/100)^2 \times 100$, so that the benchmark state scores 100. The parabola is not arbitrary. We show that it follows uniquely from three weak axioms—scale-free normalization, homogeneity of degree two, and monotone continuity—and we defend the degree-two axiom on economic grounds: pairwise interaction channels compound quadratically, agenda power is convex in coverage, and fixed costs of representation create scale economies. The resulting theory delivers sharp, testable consequences: a twelve-member "half-power club" whose members hold 55.8% of global diplomatic power with only 36.9% of posts; a systematic amplification of inequality (the Gini coefficient rises from 0.325 to 0.543 under the square-law transform); and a "middle-power trap" in which the median state is 2.56 times behind the leader in missions but 6.56 times behind in power.

Our empirical contribution is to apply the theory to the complete cross-section of 66 countries and entities tracked by the Lowy Institute's Global Diplomacy Index 2024 [7], as compiled on Wikipedia [11]: 7,745 posts in total, concentrated in a top-heavy hierarchy well described by a power-law (rank-size) regression with an estimated exponent of -0.635. An auxiliary econometric exercise links network size to fundamentals: mission counts scale with gross domestic product (elasticity 0.32, $t = 11.1$) but essentially not at all with population ($t = 0.9$), implying that actual diplomatic power behaves as a convex function of economic capacity, with an implied GDP elasticity of about 0.64.

The paper connects several literatures. It imports the axiomatic methodology of power indices and income-inequality measurement [1, 5] into diplomacy studies; it operationalizes, for embassy networks, the classical insight that power is the capacity to influence the behavior of others [2, 3, 8]; and it speaks to the economics of alliances, where public-good aggregation and burden sharing have long been central [9]. Section 2 describes the data. Section 3 develops the theory and its proofs. Section 4 presents the empirical analysis. Section 5 draws out geopolitical, welfare-economic, and security implications, Section 6 discusses limitations, and Section 7 concludes. Appendix A gives the computational procedure and Appendix B the full ranking of all 66 entities.

2 Data: International Diplomatic Representation

2.1 Source and coverage

Our universe is the set of 66 countries and entities covered by the Lowy Institute Global Diplomacy Index 2024 [7], retrieved from the Wikipedia list of countries by number of diplomatic missions [11]. For each entity the index records the number of embassies and high commissions, consulates and consulates-general, permanent missions to international organizations, and other representations (for example, trade or interest offices), together with the total number of posts m_i. The index is the most complete publicly available census of diplomatic infrastructure; its 66 tracked entities account for the overwhelming majority of all missions operated worldwide. We treat the European Union (143 posts) as a distinct entity: it is listed by the source but is a supranational organization rather than a state, so country-level statistics below are computed over the 65 states, while the full ranking of 66 entities appears in Appendix B.

2.2 Composition and descriptive statistics

Table 1 decomposes the 7,602 posts operated by the 65 tracked states. Embassies and high commissions—the channels of state-to-state political relations—dominate with two thirds of all posts. Consulates, which deliver citizen services and commercial promotion, account for one quarter. Permanent missions to multilateral organizations are only 5.2% of the stock, but as Section 5 argues they are disproportionately visible for their size.

Table 1: Composition of diplomatic posts across the 65 tracked states.

Mission type	Share of posts (%)
Embassies / high commissions	66.63
Consulates / consulates-general	25.18
Permanent missions	5.18
Other representations	2.99

Table 2 reports summary statistics for the cross-section of total posts. The distribution is right-skewed (skewness 0.67) with a moderately long upper tail: the mean state operates 117 posts, the median only 107, and the leading state 274. Dispersion is substantial—the coefficient of variation is 0.59 and the Gini coefficient of raw posts is 0.325—but as we shall see, this measured inequality is dramatically understated once posts are converted into power. Four of the 66 entries show component sums differing from published totals by one or two posts (rounding and reclassification in the source); we retain published totals throughout.

Table 2: Summary statistics for total diplomatic posts (65 states).

Statistic	Value
Countries / entities	65 / 66
Total posts (all entities)	7,745
Mean posts (countries)	117.0
Median posts (countries)	107
Standard deviation	68.6
Coefficient of variation	0.587
Skewness	0.674
Excess kurtosis	-0.372
Minimum / Maximum	10 / 274
Gini coefficient (IDR)	0.3245
Gini coefficient (ADP)	0.5426

2.3 From posts to representation

Raw counts conflate scale with relative standing. We therefore normalize:

Definition 1 (International Diplomatic Representation)**.** Let m_i denote the number of diplomatic posts operated by state i and let $m^* = \max_j m_j$ be the global benchmark. The *International Diplomatic Representation* of state i is

$$\mathrm{IDR}_i \;=\; 100 \cdot \frac{m_i}{m^*}, \tag{1}$$

so that the leading state scores exactly 100.

Figure 1 summarizes the measurement pipeline. In the 2024 cross-section the benchmark is China with $m^* = 274$ posts, followed by the United States at 271 (IDR = 98.9), Turkey at 252 (IDR = 92.0), Japan at 251, and France at 249. The median state scores IDR = 39.1; Bhutan, with ten posts, scores 3.7.

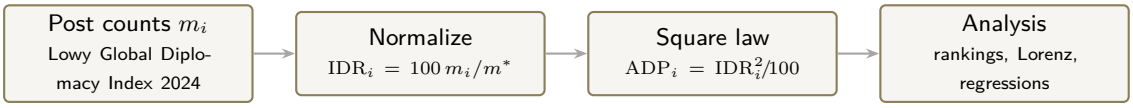


Figure 1: The measurement pipeline from raw post counts to actual diplomatic power.

3 The Theory of Actual Diplomatic Power

3.1 The square law and its axiomatic foundation

Let $x_i = m_i/m^* \in (0,1]$ denote state i's representation relative to the benchmark, and let $F(x)$ map relative representation into a power score. We impose three axioms.

Axiom 1 (Scale-free normalization)**.** F is defined on relative representation $x \in [0,1]$ and satisfies $F(1) = 100$; power is benchmark-relative and unit-free.

Axiom 2 (Homogeneity of degree two)**.** For every scaling factor $\lambda > 0$, $F(\lambda x) = \lambda^2 F(x)$: multiplying a network's relative size by λ multiplies its power by λ^2.

Axiom 3 (Monotone continuity)**.** F is continuous and strictly increasing on $[0,1]$, with $F(0) = 0$.

Theorem 1 (Characterization of actual diplomatic power)**.** *Under Axioms 1–3, F is uniquely determined:*

$$F(x) = 100\,x^2, \qquad \textit{hence} \qquad \mathrm{ADP}_i \;=\; 100\left(\frac{m_i}{m^*}\right)^2 \;=\; \frac{\mathrm{IDR}_i^2}{100}. \tag{2}$$

Proof. Set $\lambda = x$ in Axiom 2 and apply it to the point $x = 1$: $F(x) = F(x \cdot 1) = x^2 F(1) = 100\,x^2$, where the last equality is Axiom 1. Uniqueness is immediate because the identity holds for every $x \in [0,1]$. Existence is checked directly: $F(x) = 100x^2$ is continuous, strictly increasing on $[0,1]$, vanishes at zero, satisfies $F(1) = 100$, and $F(\lambda x) = 100\lambda^2 x^2 = \lambda^2 F(x)$, so all three axioms hold. □

Remark 1. The axioms are weak, but Axiom 2 carries the theory's economic content, so its defense deserves attention. Three mechanisms support degree two rather than degree one. First, *interaction compounding*: a post is valuable chiefly as a channel, and the number of distinct bilateral and trilateral interaction channels a network of m posts can support grows on the order of m^2—the same combinatorics that underlies Metcalfe-type network laws. The two-hundredth post does not merely add one relationship; it adds relationships *with every other platform* the network already operates. Second, *convex agenda power*: multilateral agenda-setting, coalition formation, and the ability to run concurrent negotiations all require simultaneous coverage; partial coverage yields disproportionally partial influence, while full coverage unlocks the whole set of venues [6, 8]. Third, *fixed costs and scale economies*: maintaining a post involves a large fixed component (premises, security, accreditation); states that cross fixed-cost thresholds in many markets simultaneously convert presence into globally coordinated action that smaller networks physically cannot mount.

3.2 Marginal power and increasing returns

The first consequence of the square law is that the marginal value of representation rises with scale.

Proposition 1 (Marginal power)**.** $\partial \mathrm{ADP}_i/\partial m_i = 200\,m_i/(m^*)^2 > 0$ *and is increasing in m_i: each additional post is worth more at larger network sizes. The elasticity of power with respect to representation is exactly* 2*.*

Proof. Differentiate $\mathrm{ADP}_i = 100\,m_i^2/(m^*)^2$ with respect to m_i. The first derivative is $200\,m_i/(m^*)^2 > 0$ and the second derivative is $200/(m^*)^2 > 0$, so marginal power is positive and increasing.

The elasticity is

$$\frac{\partial \mathrm{ADP}}{\partial m} \cdot \frac{m}{\mathrm{ADP}} = \frac{200m}{(m^*)^2} \cdot \frac{m}{100\,m^2/(m^*)^2} = 2.$$

□

Quantitatively, at China's frontier ($m = 274$) the next post is worth $\Delta\mathrm{ADP} = 100(2m+1)/(m^*)^2 \approx 0.73$ points; at the median state ($m = 107$) it is worth ≈ 0.29 points. The square law thus encodes a "Matthew effect" of diplomacy—power begets power—which is the micro-foundation of the concentration results below.

3.3 The half-power club and the middle-power trap

Proposition 2 (Half-power threshold)**.** $\mathrm{ADP}_i \geq 50$ *if and only if* $\mathrm{IDR}_i \geq 100/\sqrt{2} \approx 70.7$*, equivalently* $m_i \geq m^*/\sqrt{2} = 193.75$*. In the 2024 cross-section exactly 12 of 65 states clear the threshold—China, the United States, Turkey, Japan, France, Russia, the United Kingdom, Germany, Spain, Italy, Brazil, and India—and together they hold 55.75% of total ADP with 36.92% of total posts.*

Proof. $\mathrm{ADP}_i \geq 50 \iff (m_i/m^*)^2 \geq 1/2 \iff m_i \geq m^*/\sqrt{2}$, using strict monotonicity. The list follows from $m^* = 274$ and the ordering of the data; the shares are computed from Table 5. □

Proposition 3 (Relative power premium)**.** *If state a operates ρ times as many posts as state b, then a's power is ρ^2 times b's. Hence relative representation gaps are squared in power space.*

The proposition quantifies the *middle-power trap.* India, the twelfth-ranked network, is within 26.6% of China's post count ($\rho = 1.363$)—apparently a near-peer in presence—yet holds only 53.8 points of power against China's 100, a gap of 46.2 points. The median state is a factor of 2.56 behind China in missions but a factor of 6.56 in power. Countries in the bottom posts-quartile average an ADP score of just 2.39: in power space they are rounding errors. Figure 2 visualizes the parabola, the half-power threshold, and the position of the median state.

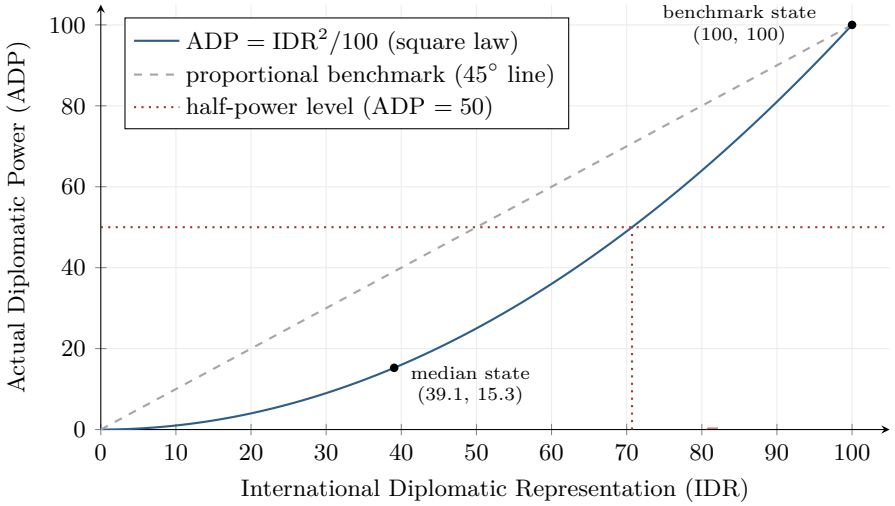


Figure 2: The parabolic power map. Above the 45° line the square law awards power premia; only the region $\mathrm{IDR} \geq 100/\sqrt{2}$ reaches half of the benchmark's power.

3.4 Concentration amplification

Because the square law is convex, it must make the power distribution strictly more unequal than the representation distribution. This is the theory's central distributional theorem.

Theorem 2 (Lorenz dominance of power)**.** *Let $\{x_1 \leq \cdots \leq x_n\}$ be positive relative representations with $x_i < 1$ for at least one $i < n$, and let $y_i = x_i^2$. Let L_X and L_Y be the Lorenz curves of $\{x_i\}$ and $\{y_i\}$. Then $L_Y(p) \leq L_X(p)$ for every $p \in [0,1]$, with strict inequality for all interior p; the power distribution Lorenz-dominates the representation distribution.*

Proof. Fix $k < n$ and let $S_k = \sum_{i \leq k} x_i$, $S = \sum_{i=1}^n x_i$, $Q_k = \sum_{i \leq k} x_i^2$, $Q = \sum_{i=1}^n x_i^2$. We must show $Q_k/Q \leq S_k/S$, i.e. $S \cdot Q_k \leq S_k \cdot Q$. Write $Q_k/S_k = \mu_k$ and $Q/S = \mu$, the x-weighted means of x over the bottom-k set and over the whole population respectively. Each such weighted mean is a convex combination of the values x_j whose weights increase with x_j itself; over the bottom-k set all values are at most x_k, so $\mu_k \leq x_k$. The whole-population mean μ is the weighted average of μ_k and the weighted mean μ_{-k} of the complement, where every value in the complement is at least $x_k \geq \mu_k$; hence $\mu_{-k} \geq \mu_k$ and therefore $\mu \geq \mu_k$, i.e. $Q_k/Q \leq S_k/S$. Strictness for interior p

follows because the complement is nonempty and contains a value strictly above x_k, so $\mu_{-k} > \mu_k$ whenever $k < n$. Dividing cumulative sums by S and Q gives $L_Y(p) \leq L_X(p)$ pointwise. □

Corollary 1 (Gini amplification)**.** *Under the square law the Gini coefficient of power is at least the Gini coefficient of representation:* $G(\text{ADP}) \geq G(\text{IDR})$*, with equality only in degenerate cases [1, 5].*

In the data the amplification is dramatic: $G(\text{IDR}) = 0.3245$ rises to $G(\text{ADP}) = 0.5426$, a 67% increase (Table 2 and Figure 3). The top five states hold 17.1% of the world's posts but 28.3% of its diplomatic power; the top ten, 31.6% and 48.8%; the top twenty, 53.0% and 71.5%. Diplomatic power is thus roughly twice as concentrated as the diplomatic infrastructure from which it derives.

4 Empirical Analysis

4.1 Rankings

Table 3 reports the top 25 of the full ranking; Appendix B lists all 66 entities. Three features stand out. First, the top two are effectively tied in representation (IDR of 100.0 versus 98.9) and remain tied in power (100.0 versus 97.8): at the very top of the distribution the parabola is locally flat, so marginal differences in posts barely move power. Second, power losses per point of representation lost are steepest at the top of the hierarchy: the drop from rank 2 to rank 7 forfeits 16.8 points of IDR but 30.4 of ADP (average slope 1.8), whereas the comparable drop from rank 10 to rank 15 forfeits 17.9 points of IDR for only 23.7 of ADP (average slope 1.3)—a direct observation of the parabola steepening in representation space. Third, several states punch visibly above their economic weight in representation (Turkey, Japan, France), a point we return to in Section 5.

Table 3: Top 25 states by total posts, with normalized representation (IDR) and actual diplomatic power (ADP). Benchmark: China, $m^* = 274$.

Rank	State	Posts m	Emb.	Cons.	IDR	ADP
1	China	274	173	91	100.0	100.00
2	United States	271	168	83	98.9	97.82
3	Turkey	252	145	93	92.0	84.59
4	Japan	251	152	66	91.6	83.92
5	France	249	158	72	90.9	82.58
6	Russia	230	143	74	83.9	70.46
7	United Kingdom	225	156	51	82.1	67.43
8	Germany	225	154	57	82.1	67.43
9	Spain	218	116	91	79.6	63.30
10	Italy	206	124	74	75.2	56.52
11	Brazil	205	135	58	74.8	55.98
12	India	201	142	50	73.4	53.81
13	South Korea	187	114	60	68.2	46.58
14	Mexico	161	80	71	58.8	34.53
15	Canada	157	98	38	57.3	32.83
16	Argentina	150	87	54	54.7	29.97
17	Netherlands	149	106	28	54.4	29.57
18	European Union	143	132	0	52.2	27.24
19	Switzerland	141	102	30	51.5	26.48
20	Hungary	140	87	43	51.1	26.11
21	Poland	135	91	33	49.3	24.28

Continued on next page

Table 3 (continued)

Rank	State	Posts m	Emb.	Cons.	IDR	ADP
22	Greece	134	85	39	48.9	23.92
23	Indonesia	130	92	34	47.4	22.51
24	Saudi Arabia	128	104	19	46.7	21.82
25	Portugal	127	75	42	46.4	21.48

4.2 Concentration: representation versus power

Figure 3 plots the Lorenz curves of IDR and ADP against the equality diagonal. The power curve bows visibly deeper: the bottom half of states, which together operate 26.4% of the world's missions, generate 12.5% of its diplomatic power. The empirical Lorenz curves thus confirm Theorem 2 quantitatively, and the 1.67× Gini amplification provides a single-number summary of how the square law reshapes the hierarchy.

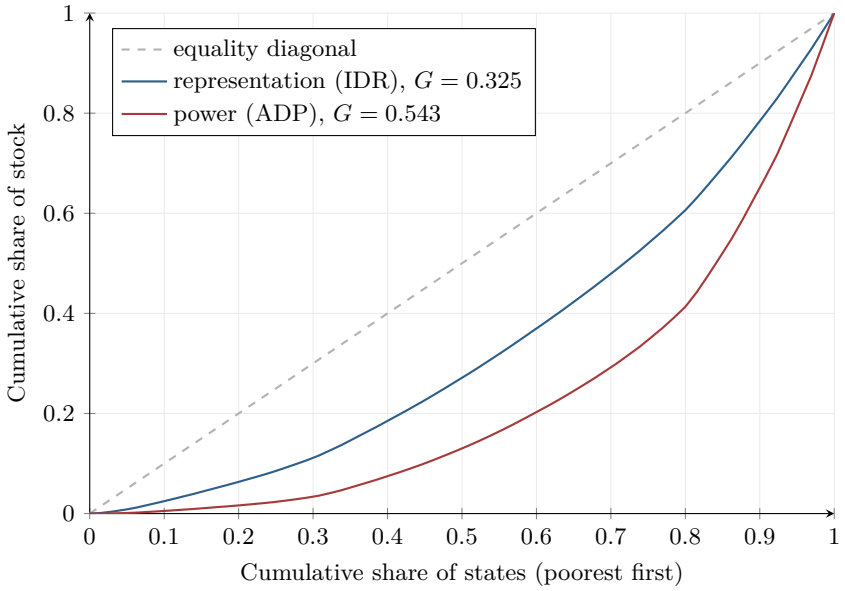


Figure 3: Lorenz curves of international diplomatic representation and actual diplomatic power across the 65 tracked states. The square law (Theorem 2) pushes the power curve everywhere below the representation curve.

4.3 The rank-size structure of diplomatic networks

Because the square law is an axiom rather than a hypothesis, estimation plays a different role here than in a conventional empirical model: it characterizes the *representation* distribution that feeds the parabola, and the *fundamentals* that determine it. We estimate two cross-sectional regressions (Table 4).

Column (1) fits the classical rank-size (Zipf) specification [12, 4]

$$\log m_{(r)} = \alpha + \beta \log r + \varepsilon_r, \tag{3}$$

where $m_{(r)}$ is the post count at rank r. The estimate is $\hat{\beta} = -0.635$ with standard error 0.052 ($t = -12.2$) and $R^2 = 0.70$: diplomatic networks decay with rank more slowly than the unit exponent of Zipf's law, reflecting an upper tier of genuinely global networks followed by a dense middle class of regional actors. Figure 4 overlays the fitted curve on the raw cross-section.

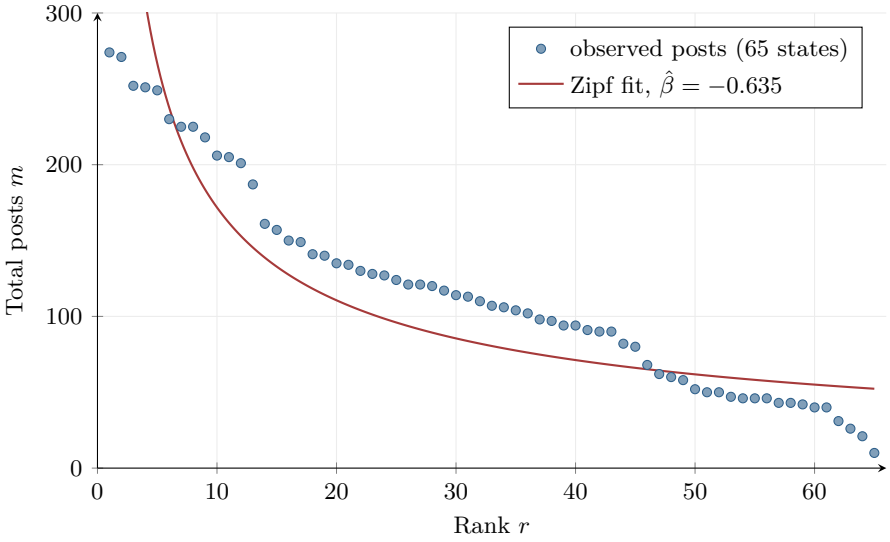


Figure 4: Rank–size distribution of diplomatic networks and the fitted power law of equation (3).

4.4 Fundamentals: what determines network size?

Column (2) of Table 4 regresses the logarithm of post counts on the logarithms of GDP and population (World Bank data; $n = 63$ states with complete data, 2023 values, most recent available):

$$\log m_i = \beta_0 + \beta_1 \log \mathrm{GDP}_i + \beta_2 \log \mathrm{POP}_i + u_i. \tag{4}$$

Economic size dominates: the GDP elasticity is 0.320 ($t = 11.1$), while the population elasticity is a statistically insignificant 0.026 ($t = 0.9$); the model explains 84.4% of the variance. Diplomatic networks, in short, are purchased by fiscal capacity, not by demographic scale—a state of 1.4 billion people and one of 60 million with the same GDP maintain networks of the same size. Combining the two estimates yields the theory's macro bridge: since ADP $\propto m^2$ and $m \propto \mathrm{GDP}^{0.32}$, actual diplomatic power behaves in the cross-section as ADP $\propto \mathrm{GDP}^{0.64}$ (implied standard error ≈ 0.06). Economic size translates into diplomatic power more than one-for-one but less than quadratically—convex, yet bounded by the budget constraint that funds each post.

Table 4: Cross-sectional regressions. Dependent variable: log posts. Column (1): rank–size (Zipf) specification over 65 states. Column (2): fundamentals specification with World Bank GDP and population, 2023. Standard errors in parentheses.

	(1) Zipf	(2) Footprint
log rank	-0.6352	
	(0.0521)	
log GDP		0.3196
		(0.0287)
log population		0.0258
		(0.0296)
Constant	6.6089	-4.4076
	(0.1742)	(0.5105)
Observations	65	63
R^2	0.7024	0.8442
Adjusted R^2	0.6977	0.8390

5 Geopolitical and Economic Implications

5.1 The geography of the half-power club

The twelve members of the half-power club (Proposition 2) are the only states whose networks independently generate global reach, and the club's internal structure is itself skewed: the top four members alone—China, the United States, Turkey, and Japan—account for 23.1% of global ADP, some 41.5% of the club's own power. For alliance planners the implication is stark—any coalition that aspires to worldwide diplomatic action must include at least one club member, and coalitional reach is priced by the square law, not the arithmetic of post counts.

5.2 Aggregation, alliances, and the EU paradox

Proposition 4 (Superadditivity of pooled representation)**.** *If states a and b pool their posts into a single network of $m_a + m_b$ posts while the benchmark m^* is held fixed, pooled power exceeds the sum of standalone powers:*

$$\frac{100\,(m_a + m_b)^2}{(m^*)^2} - \left(\frac{100\,m_a^2}{(m^*)^2} + \frac{100\,m_b^2}{(m^*)^2}\right) = \frac{200\,m_a m_b}{(m^*)^2} > 0, \tag{5}$$

and the multiplicative gain is $1 + 2m_a m_b/(m_a^2 + m_b^2) \in (1, 2]$.

Pooling is therefore superadditive under the square law: jointly, France and Germany's 474 posts would generate roughly twice ($\times 1.99$) the power the two networks generate separately. The European Union makes the point empirically vivid. As a single institutional actor the EU operates 143 posts, scoring ADP = 27.2—a middle power. But the 22 EU member states tracked in the index operate 2,641 posts among them, holding 34.7% of the world's missions and a combined standalone power equal to 5.21 times the benchmark's score. The same stock of diplomatic infrastructure therefore "counts" nineteen times more powerfully (521.2/27.2) when deployed by twenty-two sovereigns than by one institution—provided, and only provided, the twenty-two coordinate their deployment. The EU paradox is the square law's most consequential policy lesson: *coordination, not construction, is the binding constraint on European diplomatic power* [9].

5.3 Welfare economics of diplomatic networks

If a diplomatic network produces global public goods—early warning, conflict mediation, consular protection, coordination of transnational policy—then the square law implies a systematic divergence between private and social incentives. A state contemplating the marginal post of Proposition 1 internalizes its own convex return, but the mediation and crisis-prevention services a large network supplies benefit foreigners as well, so small and middle states under-provide relative to the global optimum and free-ride on the networks of the great powers [9]. Conversely, the convexity that punishes middle powers also makes *pooled provision* through alliances spectacularly productive (Proposition 4), giving welfare economics a precise argument for burden-shared, jointly commanded diplomatic infrastructure.

5.4 The economics of coercion and signaling

The theory also prices diplomatic statecraft in crises. Because power is convex in representation, withdrawing or downgrading missions is a *costly signal*: the sender forgoes a marginal product that rises with its own network size, which is precisely why embassy drawdowns are credible indicators of resolve [2]. For small states the same convexity cuts the other way—hosting ratios matter as much as sending ratios, since a state that hosts many foreign posts embeds itself in others' networks at no convex cost of its own, converting geography into borrowed representation.

6 Limitations and Extensions

The theory is deliberately parsimonious, and its limits mark the path for extension. First, the square law treats all posts as homogeneous: an embassy in a pivotal capital and a consulate in a resort town carry equal weight, though a quality-weighted representation index $m_i = \sum_k w_k n_{ik}$ would slot directly into equation (2). Second, posts measure presence, not performance; staff numbers, budgets, and language capability are unobserved. Third, the benchmark normalization ties all scores to a single leader, so idiosyncratic expansion by the benchmark state (China's network grew rapidly over the last decade) mechanically compresses everyone else's scores; a fixed historical benchmark or a panel of benchmarks is a natural robustness check. Fourth, hosted missions enter other states' networks but not the host's score, although hosting generates influence; a symmetric two-mode network model would capture this. Fifth, the GDP–population regression is descriptive rather than causal—fiscal capacity and foreign-policy ambition are jointly determined—so the implied power elasticity of GDP should be read as an accounting relationship, not a structural estimate. Finally, the index tracks 66 entities; extending the census to all UN members would mostly add small networks, which the square law penalizes, and would therefore if anything *strengthen* the measured concentration.

7 Conclusion

This paper proposed and tested a simple theory in which actual diplomatic power is a parabolic function of international diplomatic representation. The square law follows from three weak axioms; it explains why the top twelve networks—with barely more than a third of the world's posts—command a majority of its diplomatic power; it amplifies measured inequality by a factor of 1.67; it prices alliance coordination (the EU's nineteen-fold aggregation gap) and middle-power strategies with the same instrument; and it links diplomatic power to economic fundamentals with a convex elasticity of roughly 0.64. Future work should quality-weight the post stock, extend the census, and exploit the panel dimension to estimate the dynamics of power transitions under the square law [10]. The broader lesson is geometrical: in diplomacy, as in networks generally, the map from inputs to power is curved—and it is curved against those in the middle.

A Computational Procedure

Algorithm 1 summarizes the pipeline of Figure 1: from raw post counts to the ADP index, Lorenz curves, and concentration statistics. All quantities used in this paper are reproducible from the public source data [7, 11] with these steps alone.

Algorithm 1: Computation of actual diplomatic power and concentration statistics.

Input: Post counts $\{m_i\}_{i=1}^{n}$ (embassies, consulates, permanent missions, other), $n = 65$ states
Output: ADP index, Lorenz curves $L_{\mathrm{IDR}}, L_{\mathrm{ADP}}$, Gini coefficients, top-k shares

$m^* \leftarrow \max_i m_i$;
for $i \leftarrow 1$ **to** n **do**
  $\mathrm{IDR}_i \leftarrow 100\, m_i / m^*$; `// Definition 1`
  $\mathrm{ADP}_i \leftarrow \mathrm{IDR}_i^2 / 100$; `// Theorem 1`
end
sort $(\mathrm{IDR}, \mathrm{ADP})$ jointly by ascending m_i into $\{(\tilde{x}_i, \tilde{y}_i)\}$;
for $k \leftarrow 0$ **to** n **do**
  $L_{\mathrm{IDR}}(k/n) \leftarrow \big(\sum_{i \le k} \tilde{x}_i\big) / \sum_i \tilde{x}_i$;
  $L_{\mathrm{ADP}}(k/n) \leftarrow \big(\sum_{i \le k} \tilde{y}_i\big) / \sum_i \tilde{y}_i$;
end
$G \leftarrow 1 - 2\int_0^1 L(p)\,\mathrm{d}p$; `// applied to each Lorenz curve`
return $\{\mathrm{ADP}_i\}$, L_{IDR}, L_{ADP}, $\{G\}$, *top-k shares*;

B Full Ranking

Table 5 lists all 66 entities with their post counts by mission type, IDR, and ADP. The European Union (rank "–") is included for completeness but excluded from state-level statistics.

Table 5: Full ranking of international diplomatic representation (IDR) and actual diplomatic power (ADP), Lowy Institute Global Diplomacy Index 2024.

Rank	State	Posts m	Emb.	Cons.	Perm.	IDR	ADP
1	China	274	173	91	8	100.0	100.00
2	United States	271	168	83	11	98.9	97.82
3	Turkey	252	145	93	12	92.0	84.59
4	Japan	251	152	66	10	91.6	83.92
5	France	249	158	72	18	90.9	82.58
6	Russia	230	143	74	10	83.9	70.46
7	United Kingdom	225	156	51	11	82.1	67.43
8	Germany	225	154	57	12	82.1	67.43
9	Spain	218	116	91	10	79.6	63.30
10	Italy	206	124	74	8	75.2	56.52
11	Brazil	205	135	58	12	74.8	55.98
12	India	201	142	50	5	73.4	53.81
13	South Korea	187	114	60	5	68.2	46.58
14	Mexico	161	80	71	8	58.8	34.53
15	Canada	157	98	38	11	57.3	32.83
16	Argentina	150	87	54	7	54.7	29.97
17	Netherlands	149	106	28	10	54.4	29.57
–	European Union	143	132	0	8	52.2	27.24
18	Switzerland	141	102	30	7	51.5	26.48
19	Hungary	140	87	43	7	51.1	26.11
20	Poland	135	91	33	9	49.3	24.28
21	Greece	134	85	39	9	48.9	23.92
22	Indonesia	130	92	34	3	47.4	22.51
23	Saudi Arabia	128	104	19	4	46.7	21.82
24	Portugal	127	75	42	9	46.4	21.48
25	Australia	124	85	33	4	45.3	20.48
26	Pakistan	121	89	29	3	44.2	19.50
27	Chile	121	73	41	6	44.2	19.50
28	Czech Republic	120	91	16	7	43.8	19.18
29	Colombia	117	63	49	5	42.7	18.23
30	South Africa	114	98	12	2	41.6	17.31
31	Belgium	113	83	18	8	41.2	17.01
32	Taiwan	110	12	1	1	40.1	16.12
33	Israel	107	80	21	5	39.1	15.25
34	Malaysia	106	78	24	3	38.7	14.97
35	Austria	104	83	13	6	38.0	14.41
36	Sweden	102	87	8	7	37.2	13.86
37	Ireland	98	71	20	6	35.8	12.79
38	Thailand	97	65	28	3	35.4	12.53
39	Vietnam	94	67	22	4	34.3	11.77
40	Philippines	94	63	27	4	34.3	11.77
41	Norway	91	74	7	9	33.2	11.03
42	Finland	90	74	6	7	32.8	10.79
43	Denmark	90	66	16	6	32.8	10.79

(continued on next page)

(continued)

Rank	State	Posts m	Emb.	Cons.	Perm.	IDR	ADP
44	Slovakia	82	63	9	8	29.9	8.96
45	Bangladesh	80	58	20	2	29.2	8.52
46	New Zealand	68	49	15	3	24.8	6.16
47	Lithuania	62	42	11	8	22.6	5.12
48	Sri Lanka	60	46	12	1	21.9	4.80
49	Slovenia	58	43	7	7	21.2	4.48
50	Costa Rica	52	42	5	4	19.0	3.60
51	Mongolia	50	33	13	1	18.2	3.33
52	Singapore	50	29	16	4	18.2	3.33
53	Latvia	47	38	1	7	17.2	2.94
54	Estonia	46	38	2	6	16.8	2.82
55	Myanmar	46	38	6	2	16.8	2.82
56	Luxembourg	46	35	4	6	16.8	2.82
57	North Korea	43	39	1	3	15.7	2.46
58	Cambodia	43	28	12	3	15.7	2.46
59	Brunei	42	35	4	3	15.3	2.35
60	Nepal	40	32	7	1	14.6	2.13
61	Laos	40	26	11	3	14.6	2.13
62	Timor-Leste	31	23	5	3	11.3	1.28
63	Iceland	26	17	4	5	9.5	0.90
64	Papua New Guinea	21	16	4	1	7.7	0.59
65	Bhutan	10	6	3	1	3.6	0.13

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